

Homework 5 - SSIE 553

Lana Jalal Gidan | Spring 2026 | Prof. Neha Patankar

Question 1(i)

Primal:

$$\text{Max } z = 2x_1 + x_2$$

s.t.

$$-x_1 + x_2 \leq 1 \quad \dots(1)$$

$$x_1 + x_2 \leq 3 \quad \dots(2)$$

$$x_1 - 2x_2 \leq 4 \quad \dots(3)$$

$$x_1, x_2 \geq 0$$

Step 1: Identify dual variables

Since all primal constraints are ($\leq$) and the objective is MAX, the dual is a MIN problem. Each $\leq$ constraint produces one dual variable ≥ 0 :

Constraint (1) is $\leq \Rightarrow y_1 \geq 0$

Constraint (2) is $\leq \Rightarrow y_2 \geq 0$

Constraint (3) is $\leq \Rightarrow y_3 \geq 0$

Step 2: Write the dual objective

The dual objective is MIN, and the RHS values of the primal constraints become the objective coefficients of the dual:

$$\text{MIN } w = 1*y_1 + 3*y_2 + 4*y_3$$

Step 3: Write dual constraints (one per primal variable)

Each primal variable $x_j \geq 0$ generates a dual constraint of the form:

$$(\text{column } j \text{ of } A)^T * y \geq c_j$$

For x_1 ($c_1 = 2$): read coefficients of x_1 in rows (1),(2),(3): -1, +1, +1

$$\Rightarrow -y_1 + y_2 + y_3 \geq 2$$

For x_2 ($c_2 = 1$): coefficients of x_2 in rows (1),(2),(3): +1, +1, -2

$$\Rightarrow y_1 + y_2 - 2y_3 \geq 1$$

DUAL PROBLEM:

$$\text{Min } w = y_1 + 3y_2 + 4y_3$$

s.t.

$$-y_1 + y_2 + y_3 \geq 2$$

$$y_1 + y_2 - 2y_3 \geq 1$$

$$y_1, y_2, y_3 \geq 0$$

Question 1(ii)

Primal:

$$\text{Min } w = y_1 - y_2$$

s.t.

$$2y_1 + y_2 \geq 4 \quad \dots(5)$$

$$y_1 + y_2 \geq 1 \quad \dots(6)$$

$$y_1 + 2y_2 \geq 3 \quad \dots(7)$$

$$y_1, y_2 \geq 0$$

Step 1: Identify dual variables

The primal is MIN, so its dual is MAX. Each $\geq$ constraint produces one dual variable ≥ 0 :

$$\text{Constraint (5) is } \geq \Rightarrow x_1 \geq 0$$

$$\text{Constraint (6) is } \geq \Rightarrow x_2 \geq 0$$

$$\text{Constraint (7) is } \geq \Rightarrow x_3 \geq 0$$

Step 2: Write the dual objective

The dual is MAX and the RHS values of the primal become the dual objective coefficients:

$$\text{MAX } z = 4x_1 + 1x_2 + 3x_3$$

Step 3: Write dual constraints (one per primal variable)

Each primal variable $y_j \geq 0$ generates a dual constraint of the form:

$$(\text{column } j \text{ of } A)^T * x \leq c_j$$

For y_1 ($c_1 = 1$): coefficients of y_1 in rows (5),(6),(7): 2, 1, 1

$$\Rightarrow 2x_1 + x_2 + x_3 \leq 1$$

For y_2 ($c_2 = -1$): coefficients of y_2 in rows (5),(6),(7): 1, 1, 2

$$\Rightarrow x_1 + x_2 + 2x_3 \leq -1$$

DUAL PROBLEM:

$$\text{Max } z = 4x_1 + x_2 + 3x_3$$

s.t.

$$2x_1 + x_2 + x_3 \leq 1$$

$$x_1 + x_2 + 2x_3 \leq -1$$

$$x_1, x_2, x_3 \geq 0$$

Question 1(iii)

Primal:

$$\text{Max } z = 4x_1 - x_2 + 2x_3$$

s.t.

$$x_1 + x_2 \leq 5 \quad \dots(9)$$

$$2x_1 + x_2 \leq 7 \quad \dots(10)$$

$$2x_2 + x_3 \geq 6 \quad \dots(11)$$

$$x_1 + x_3 = 4 \quad \dots(12)$$

$$x_1 \geq 0; \quad x_2, x_3 \text{ unrestricted (urs)}$$

Step 1: Assign dual variables

For a MAX primal, the sign of each dual variable depends on the constraint type:

$$\text{Constraint (9) is } \leq \Rightarrow y_1 \geq 0$$

$$\text{Constraint (10) is } \leq \Rightarrow y_2 \geq 0$$

$$\text{Constraint (11) is } \geq \Rightarrow y_3 \leq 0$$

$$\text{Constraint (12) is } = \Rightarrow y_4 \text{ unrestricted (urs)}$$

Step 2: Write the dual objective

$$\text{MIN } w = 5y_1 + 7y_2 + 6y_3 + 4y_4$$

Step 3: Write dual constraints

The type of dual constraint depends on the primal variable sign:

$$x_j \geq 0 \Rightarrow \text{dual constraint is } \geq$$

$$x_j \text{ urs} \Rightarrow \text{dual constraint is } =$$

For x_1 (≥ 0 , $c_1 = 4$): coefficients in rows (9),(10),(11),(12): 1, 2, 0, 1

$$\Rightarrow y_1 + 2y_2 + 0y_3 + y_4 \geq 4$$

For x_2 (urs, $c_2 = -1$): coefficients in rows (9),(10),(11),(12): 1, 1, 2, 0

$$\Rightarrow y_1 + y_2 + 2y_3 = -1$$

For x_3 (urs, $c_3 = 2$): coefficients in rows (9),(10),(11),(12): 0, 0, 1, 1

$$\Rightarrow y_3 + y_4 = 2$$

DUAL PROBLEM:

$$\text{Min } w = 5y_1 + 7y_2 + 6y_3 + 4y_4$$

s.t.

$$y_1 + 2y_2 + y_4 \geq 4$$

$$y_1 + y_2 + 2y_3 = -1$$

$$y_3 + y_4 = 2$$

$$y_1, y_2 \geq 0; y_3 \leq 0; y_4 \text{ urs}$$

Question 1(iv)

Primal:

$$\text{Min } w = 4y_1 + 2y_2 - y_3$$

s.t.

$$y_1 + 2y_2 \leq 6 \quad \dots(14)$$

$$y_1 - y_2 + 2y_3 = 8 \quad \dots(15)$$

$$y_1 \geq 0; y_2, y_3 \text{ unrestricted (urs)}$$

Step 1: Assign dual variables

Primal is MIN so dual is MAX. Dual variable sign depends on constraint type:

Constraint (14) is $\leq \Rightarrow x_1 \leq 0$

Constraint (15) is $= \Rightarrow x_2 \text{ unrestricted (urs)}$

Step 2: Write the dual objective

$$\text{MAX } z = 6x_1 + 8x_2$$

Step 3: Write dual constraints

Each primal variable y_j generates one dual constraint:

$y_j \geq 0 \Rightarrow \text{constraint} \leq c_j$

$y_j \text{ urs} \Rightarrow \text{constraint} = c_j$

For $y_1 (\geq 0, c_1 = 4)$: coefficients in rows (14),(15): 1, 1

$$\Rightarrow x_1 + x_2 \leq 4$$

For $y_2 (\text{urs}, c_2 = 2)$: coefficients in rows (14),(15): 2, -1

$$\Rightarrow 2x_1 - x_2 = 2$$

For $y_3 (\text{urs}, c_3 = -1)$: coefficients in rows (14),(15): 0, 2

$$\Rightarrow 2x_2 = -1$$

DUAL PROBLEM:

$$\text{Max } z = 6x_1 + 8x_2$$

s.t.

$$x_1 + x_2 \leq 4$$

$$2x_1 - x_2 = 2$$

$$2x_2 = -1$$

$$x_1 \leq 0; x_2 \text{ urs}$$

Question 2

Primal:

$$\text{Max } z = 2x_1 + 3x_2 + 6x_3$$

s.t.

$$x_1 + x_3 \leq 3 \quad \dots(19)$$

$$x_2 + x_3 \leq 5 \quad \dots(20)$$

$$x_1, x_2, x_3 \geq 0$$

Part (i): Construct the Dual

Step 1: Assign dual variables

Constraint (19) is $\leq \Rightarrow y_1 \geq 0$

Constraint (20) is $\leq \Rightarrow y_2 \geq 0$

Step 2: Dual objective

$$\text{MIN } w = 3y_1 + 5y_2$$

Step 3: Dual constraints

For x_1 ($c_1 = 2$): coefficients in (19),(20): 1, 0 $\Rightarrow y_1 \geq 2$

For x_2 ($c_2 = 3$): coefficients in (19),(20): 0, 1 $\Rightarrow y_2 \geq 3$

For x_3 ($c_3 = 6$): coefficients in (19),(20): 1, 1 $\Rightarrow y_1 + y_2 \geq 6$

DUAL PROBLEM:

$$\text{Min } w = 3y_1 + 5y_2$$

s.t.

$$y_1 \geq 2$$

$$y_2 \geq 3$$

$$y_1 + y_2 \geq 6$$

$$y_1, y_2 \geq 0$$

Part (ii): Solve the Dual Graphically

Step 1: Identify constraint boundary lines in the y_1 - y_2 plane

L1: $y_1 = 2$ (vertical line; feasible region is to the right: $y_1 \geq 2$)

L2: $y_2 = 3$ (horizontal line; feasible region is above: $y_2 \geq 3$)

L3: $y_1 + y_2 = 6$ (diagonal line; feasible region is above: $y_1 + y_2 \geq 6$)

Note: if $y_1 \geq 2$ and $y_2 \geq 3$, then $y_1 + y_2 \geq 5$, but we need ≥ 6 .

So constraint L3 is active near the lower-left part of the feasible region and is NOT redundant.

Step 2: Find the corner (extreme) points of the feasible region

Corner A: intersection of L1 ($y_1=2$) and L3 ($y_1+y_2=6$):

$$y_1 = 2, \quad 2 + y_2 = 6 \Rightarrow y_2 = 4$$

$$\Rightarrow \text{Corner A} = (y_1, y_2) = (2, 4)$$

Corner B: intersection of L2 ($y_2=3$) and L3 ($y_1+y_2=6$):

$$y_2 = 3, \quad y_1 + 3 = 6 \Rightarrow y_1 = 3$$

$$\Rightarrow \text{Corner B} = (y_1, y_2) = (3, 3)$$

Beyond these two corners the feasible region continues to the upper-right (larger y_1 and y_2), but the objective $\text{MIN } 3y_1+5y_2$ increases in that direction, so the minimum must lie at one of these two corners.

Step 3: Evaluate the objective at each corner

Objective: $w = 3y_1 + 5y_2$

$$\text{Corner A} = (2, 4): \quad w = 3(2) + 5(4) = 6 + 20 = 26$$

$$\text{Corner B} = (3, 3): \quad w = 3(3) + 5(3) = 9 + 15 = 24$$

Minimum $w = 24$ occurs at Corner B = (3, 3).

DUAL OPTIMAL SOLUTION:

$$y_1^* = 3, \quad y_2^* = 3, \quad w^* = 24$$

Shadow prices of the primal constraints:

$$\text{Shadow price of constraint (19) } [x_1 + x_3 \leq 3] = y_1^* = 3$$

$$\text{Shadow price of constraint (20) } [x_2 + x_3 \leq 5] = y_2^* = 3$$

Interpretation: Each 1-unit increase in the RHS of constraint (19) or (20) increases the optimal primal profit by 3 units.

Verification via Strong Duality:

Primal solution: $x_1=0, x_2=2, x_3=3$

$$z = 2(0) + 3(2) + 6(3) = 0 + 6 + 18 = 24 = w^* \quad [\text{OK}]$$

Question 3

Primal:

$$\text{Max } z = x_1 + 2x_2$$

s.t.

$$-x_1 + x_2 \leq -2 \quad \dots(24)$$

$$4x_1 + x_2 \leq 4 \quad \dots(25)$$

$$x_1, x_2 \geq 0$$

Part (i): Show the primal has no feasible solution (graphically)

Step 1: Rewrite constraints in slope-intercept form

$$\text{Constraint (24): } -x_1 + x_2 \leq -2 \Rightarrow x_2 \leq x_1 - 2$$

$$\text{Constraint (25): } 4x_1 + x_2 \leq 4 \Rightarrow x_2 \leq 4 - 4x_1$$

$$\text{Non-negativity: } x_1 \geq 0, x_2 \geq 0$$

Step 2: Derive the implied lower bound on x_1

From (24) and $x_2 \geq 0$:

$$0 \leq x_2 \leq x_1 - 2$$

$$\text{This requires } x_1 - 2 \geq 0 \Rightarrow x_1 \geq 2$$

Step 3: Substitute $x_1 \geq 2$ into constraint (25)

$$\text{From (25): } x_2 \leq 4 - 4x_1$$

$$\text{If } x_1 \geq 2: 4 - 4x_1 \leq 4 - 4(2) = -4$$

$$\text{So } x_2 \leq -4$$

But $x_2 \geq 0$ is required. CONTRADICTION: x_2 cannot be both ≥ 0 and ≤ -4 .

CONCLUSION:

The primal feasible region is EMPTY. Graphically, constraint (24) with $x_2 \geq 0$ forces $x_1 \geq 2$, but constraint (25) then forces $x_2 \leq -4$, which violates $x_2 \geq 0$. The two half-planes do not intersect in the first quadrant.

The primal has NO FEASIBLE SOLUTION.

Part (ii): Construct the Dual

Step 1: Assign dual variables

$$\text{Constraint (24) is } \leq \Rightarrow y_1 \geq 0$$

$$\text{Constraint (25) is } \leq \Rightarrow y_2 \geq 0$$

Step 2: Dual objective

$$\text{MIN } w = -2y_1 + 4y_2$$

Step 3: Dual constraints

For x_1 ($c_1 = 1$): coefficients in (24),(25): $-1, 4 \Rightarrow -y_1 + 4y_2 \geq 1$

For x_2 ($c_2 = 2$): coefficients in (24),(25): $1, 1 \Rightarrow y_1 + y_2 \geq 2$

DUAL PROBLEM:

$$\text{Min } w = -2y_1 + 4y_2$$

s.t.

$$-y_1 + 4y_2 \geq 1$$

$$y_1 + y_2 \geq 2$$

$$y_1, y_2 \geq 0$$

Part (iii): Show the dual is unbounded (graphically)

Step 1: Confirm the dual is feasible

Try $(y_1, y_2) = (1, 1)$:

$$-1 + 4(1) = 3 \geq 1 \text{ [OK]}$$

$$1 + 1 = 2 \geq 2 \text{ [OK]}$$

$$y_1, y_2 \geq 0 \text{ [OK]}$$

The dual is feasible. $w = -2(1) + 4(1) = 2$.

Step 2: Construct a direction along which $w \rightarrow -\infty$

Let $y_1 \rightarrow +\infty$ and set $y_2 = (1 + y_1)/4$ (satisfying constraint (i) with equality).

Evaluate the objective along this ray:

$$w = -2y_1 + 4 * (1 + y_1)/4$$

$$= -2y_1 + 1 + y_1$$

$$= -y_1 + 1$$

As $y_1 \rightarrow +\infty$: $w \rightarrow -\infty$

Check constraint (ii) along this ray:

$$y_1 + y_2 = y_1 + (1+y_1)/4 = (5y_1 + 1)/4 \rightarrow +\infty \geq 2 \text{ [OK]}$$

Check $y_1, y_2 \geq 0$: $y_2 = (1+y_1)/4 > 0$ [OK]

All constraints remain satisfied as $y_1 \rightarrow +\infty$, yet $w \rightarrow -\infty$.

CONCLUSION:

The dual is UNBOUNDED ($w \rightarrow -\infty$ along the feasible ray $y_1 \rightarrow \infty$, $y_2 = (1+y_1)/4$).

This is consistent with LP Duality Theory:

Primal INFEASIBLE $\Rightarrow$ Dual is either INFEASIBLE or UNBOUNDED.
Here the dual is feasible but unbounded, confirming the primal
has no feasible solution.

Question 4

Primal:

$$\text{Max } z = 2x_1 + 5x_2 + 4x_3$$

s.t.

$$x_1 + 2x_2 + 4x_3 \leq 10 \quad \dots(29)$$

$$3x_1 + 3x_2 + 2x_3 \leq 10 \quad \dots(30)$$

$$x_1, x_2, x_3 \geq 0$$

Part (i): Construct the Dual

Step 1: Assign dual variables

Constraint (29) is $\leq \Rightarrow y_1 \geq 0$

Constraint (30) is $\leq \Rightarrow y_2 \geq 0$

Step 2: Dual objective

$$\text{MIN } w = 10y_1 + 10y_2$$

Step 3: Dual constraints

For x_1 ($c_1 = 2$): coefficients in (29),(30): 1, 3 $\Rightarrow y_1 + 3y_2 \geq 2$

For x_2 ($c_2 = 5$): coefficients in (29),(30): 2, 3 $\Rightarrow 2y_1 + 3y_2 \geq 5$

For x_3 ($c_3 = 4$): coefficients in (29),(30): 4, 2 $\Rightarrow 4y_1 + 2y_2 \geq 4$

DUAL PROBLEM:

$$\text{Min } w = 10y_1 + 10y_2$$

s.t.

$$y_1 + 3y_2 \geq 2$$

$$2y_1 + 3y_2 \geq 5$$

$$4y_1 + 2y_2 \geq 4$$

$$y_1, y_2 \geq 0$$

Part (ii): Weak Duality - Show Optimal Primal Value Cannot Exceed 25

Weak Duality Theorem states:

For any primal feasible x and any dual feasible y :

$$z = c^T x \leq b^T y = w$$

Therefore, the optimal primal value $z^* \leq$ value of ANY feasible dual solution.

We find a feasible dual solution with $w = 25$.

Try $(y_1, y_2) = (2.5, 0)$:

$$\text{Constraint 1: } y_1 + 3y_2 = 2.5 + 0 = 2.5 \geq 2 \quad [\text{OK}]$$

Constraint 2: $2y_1 + 3y_2 = 5.0 + 0 = 5.0 \geq 5$ [OK]

Constraint 3: $4y_1 + 2y_2 = 10.0 + 0 = 10 \geq 4$ [OK]

$y_1, y_2 \geq 0$: $2.5 \geq 0, 0 \geq 0$ [OK]

This point (2.5, 0) is dual feasible.

Dual objective value: $w = 10(2.5) + 10(0) = 25$

By Weak Duality: $z^* \leq w = 25$

Therefore, the optimal primal objective value cannot exceed 25.

Part (iii): Basic Solution with x_2, x_3 Basic; Complementary Slackness

Step 1: Set up the basis matrix $B = [A_2 \mid A_3]$

Non-basic variable: $x_1 = 0$.

Basis matrix formed from columns of x_2 and x_3 in the constraint matrix:

$A_2 = \text{column of } x_2 = [2, 3]^T$ (from constraints (29) and (30))

$A_3 = \text{column of } x_3 = [4, 2]^T$

$$B = \begin{bmatrix} 2 & 4 \\ 3 & 2 \end{bmatrix}$$

$$\det(B) = (2)(2) - (4)(3) = 4 - 12 = -8$$

Step 2: Compute B_{inverse}

For a 2x2 matrix $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$:

$$B_{\text{inv}} = (1/\det) * \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$B_{\text{inv}} = (1 / -8) * \begin{bmatrix} 2 & -4 \\ -3 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} -1/4 & 1/2 \\ 3/8 & -1/4 \end{bmatrix}$$

Step 3: Compute basic variable values $x_B = B_{\text{inv}} * b$, where $b = [10, 10]^T$

$$x_2 = (-1/4)(10) + (1/2)(10) = -2.5 + 5.0 = 2.5$$

$$x_3 = (3/8)(10) + (-1/4)(10) = 3.75 - 2.5 = 1.25$$

Full primal basic solution:

$$x_1 = 0, x_2 = 2.5, x_3 = 1.25$$

Primal objective value:

$$z = 2(0) + 5(2.5) + 4(1.25) = 0 + 12.5 + 5.0 = 17.5$$

Step 4: Derive the dual solution $y^* = c_B * B_{inv}$

$c_B = [c_2, c_3] = [5, 4]$ (objective coefficients of the basic variables)

$$\begin{aligned} y_1^* &= c_B * (\text{column 1 of } B_{inv}) \\ &= 5(-1/4) + 4(3/8) \\ &= -5/4 + 12/8 = -10/8 + 12/8 = 2/8 = 0.25 \end{aligned}$$

$$\begin{aligned} y_2^* &= c_B * (\text{column 2 of } B_{inv}) \\ &= 5(1/2) + 4(-1/4) \\ &= 5/2 - 1 = 3/2 = 1.5 \end{aligned}$$

Dual solution: $y_1^* = 0.25$, $y_2^* = 1.5$

Dual objective: $w = 10(0.25) + 10(1.5) = 2.5 + 15 = 17.5$

Step 5: Check Complementary Slackness (CS) conditions

CS Condition A - Primal: $y_i^* * (\text{primal slack of constraint } i) = 0$

Constraint (29): $x_1 + 2x_2 + 4x_3 = 0 + 5 + 5 = 10 \Rightarrow \text{slack} = 10 - 10 = 0$
 $y_1^* = 0.25 > 0$ and $\text{slack} = 0$: $0.25 * 0 = 0$ [OK]

Constraint (30): $3x_1 + 3x_2 + 2x_3 = 0 + 7.5 + 2.5 = 10 \Rightarrow \text{slack} = 0$
 $y_2^* = 1.5 > 0$ and $\text{slack} = 0$: $1.5 * 0 = 0$ [OK]

CS Condition B - Dual: $x_j^* * (\text{dual slack of constraint } j) = 0$

Dual constraint for x_1 : $y_1 + 3y_2 \geq 2$
LHS = $0.25 + 3(1.5) = 0.25 + 4.5 = 4.75 \Rightarrow \text{dual slack} = 4.75 - 2 = 2.75$
 $x_1^* = 0 \Rightarrow 0 * 2.75 = 0$ [OK]

Dual constraint for x_2 : $2y_1 + 3y_2 \geq 5$
LHS = $2(0.25) + 3(1.5) = 0.5 + 4.5 = 5.0 \Rightarrow \text{dual slack} = 5 - 5 = 0$
 $x_2^* = 2.5 > 0$ and $\text{dual slack} = 0$: $2.5 * 0 = 0$ [OK]

Dual constraint for x_3 : $4y_1 + 2y_2 \geq 4$

$$\text{LHS} = 4(0.25) + 2(1.5) = 1.0 + 3.0 = 4.0 \Rightarrow \text{dual slack} = 4 - 4 = 0$$
$$x_3^* = 1.25 > 0 \text{ and dual slack} = 0: 1.25 * 0 = 0 \text{ [OK]}$$

CONCLUSION:

All complementary slackness conditions are satisfied.

Since $z = w = 17.5$ (primal objective equals dual objective),
by the Strong Duality Theorem, BOTH solutions are OPTIMAL:

Primal optimal: $x_1=0, x_2=2.5, x_3=1.25 \Rightarrow z^* = 17.5$

Dual optimal: $y_1=0.25, y_2=1.5 \Rightarrow w^* = 17.5$

Question 5

LP:

$$\text{Max } z = 3x_1 + 7x_2 + 5x_3$$

s.t.

$$x_1 + x_2 + x_3 \leq 50 \quad (\text{Sugar capacity})$$

$$2x_1 + 3x_2 + x_3 \leq 100 \quad (\text{Chocolate capacity})$$

$$x_1, x_2, x_3 \geq 0$$

Optimal basis: $\{x_3, x_2\}$ (ordered: x_3 first, x_2 second)

$$\begin{vmatrix} 3/2 & -1/2 \end{vmatrix}$$

$$B_{\text{inv}} = \begin{vmatrix} -1/2 & 1/2 \end{vmatrix}$$

Part (i): Shadow prices for each constraint

Step 1: Identify basis columns and verify B_{inv}

The basis is ordered $\{x_3, x_2\}$. Reading the constraint matrix column by column:

$$A_3 = \text{column of } x_3 = [1, 1]^T \quad (\text{coefficient 1 in sugar, 1 in chocolate})$$

$$A_2 = \text{column of } x_2 = [1, 3]^T \quad (\text{coefficient 1 in sugar, 3 in chocolate})$$

$$\begin{vmatrix} 1 & 1 \end{vmatrix}$$

$$B = \begin{vmatrix} 1 & 3 \end{vmatrix}$$

$$\det(B) = (1)(3) - (1)(1) = 3 - 1 = 2$$

$$B_{\text{inv}} = (1/2) * \begin{bmatrix} 3 & -1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 3/2 & -1/2 \\ -1/2 & 1/2 \end{bmatrix}$$

This matches the given B_{inv} . [OK]

Step 2: Compute shadow prices $y^* = c_B * B_{\text{inv}}$

$$c_B = [c_3, c_2] = [5, 7] \quad (\text{objective coefficients in basis order } x_3, x_2)$$

Shadow price of the Sugar constraint (y_1^*):

$$y_1^* = c_B * (\text{column 1 of } B_{\text{inv}})$$

$$= 5(3/2) + 7(-1/2)$$

$$= 15/2 - 7/2 = 8/2 = 4$$

Shadow price of the Chocolate constraint (y_2^*):

$$\begin{aligned}
y_2^* &= c_B * (\text{column 2 of } B_{\text{inv}}) \\
&= 5(-1/2) + 7(1/2) \\
&= -5/2 + 7/2 = 2/2 = 1
\end{aligned}$$

SHADOW PRICES:

$$\begin{aligned}
y_1^* &= 4 \text{ cents per oz (Sugar)} \\
y_2^* &= 1 \text{ cent per oz (Chocolate)}
\end{aligned}$$

Interpretation:

Sugar ($y_1^* = 4$): Each additional 1 oz of sugar beyond the current 50 oz limit increases the company's optimal total profit by 4 cents, at the current optimal basis. Sugar is the more valuable resource here.

Chocolate ($y_2^* = 1$): Each additional 1 oz of chocolate beyond the current 100 oz limit increases total profit by 1 cent at the current optimal basis. Chocolate is less binding than sugar at the optimum.

Note: These shadow prices are valid only while the current basis remains optimal (within the allowable range of RHS changes).

Part (ii): Maximum willingness-to-pay for 1 extra oz of Sugar

The shadow price $y_1^* = 4$ is the marginal value of one additional oz of sugar. If the company pays a price P per extra oz of sugar, the net gain is:

$$\text{Net gain} = \text{marginal value} - \text{cost} = 4 - P$$

The company is willing to pay as long as it does not lose money:

$$4 - P \geq 0 \Rightarrow P \leq 4$$

At $P = 4$ the company breaks even; at $P > 4$ it loses money.

ANSWER:

Maximum willingness-to-pay for 1 extra oz of Sugar = 4 cents per oz.

Part (iii): Maximum willingness-to-pay for 1 extra oz of Chocolate

The shadow price $y_2^* = 1$ is the marginal value of one additional oz of chocolate. If the company pays a price P per extra oz of chocolate, the net gain is:

$$\text{Net gain} = 1 - P$$

The company is willing to pay as long as:

$$1 - P \geq 0 \Rightarrow P \leq 1$$

ANSWER:

Maximum willingness-to-pay for 1 extra oz of Chocolate = 1 cent per oz.